

أسئلة اختبارات لمادة أمن المعلومات ITI410

Question 1

You prepare an Internet usage policy for an enterprise. What is the correct statement among the following statements?

Answers

- 1- The user is permitted to surf the Internet with no restriction.
- 2- The user is permitted to surf the Internet after business work time.
- 3- The user is permitted to surf the Internet during certain period of the day.
- 4- **The user is permitted to surf the Internet for business matters only.**

Question 2

Integrity in Information System security for sending data over a network means:

Answers

- 1- **The received data is identical to the sent data.**
- 2- The received data has not been exposed to other parties.
- 3- The data sender is the one who pretend to be.
- 4- The data reach only the intended destination and not other destinations.

Question 3

What is the correct statement that describes non-repudiation:

Answers

- 1- The sender must not share their certificate with anyone.
- 2- The recipient must have a digital certificate.
- 3- **The sender cannot credibly deny that the message was sent by him.**
- 4- The recipients cannot credibly deny that they received the message.

Question 4

Which of the following is strong password?

Answers

- 1- c0mputer
- 2- My dog name
- 3- secret22
- 4- **\$jelF2bb**

Question 5

Which of the following is true about security concepts?

Answers

- 1- **Availability is related to execute action.**
- 2- DOS attack affects mainly the confidentiality of the system.
- 3- Integrity is related to read action.
- 4- All of the above.

Question 6

Which of the following is not a Risk Analysis approach:

Answers

- 1- Informal Approach
- 2- **Formal Approach**
- 3- Detailed Approach
- 4- Baseline Approach

Question 7

If the probability of facing the risk is low and the cost of the loss is high we:

Answers

- 1- Terminate the risk
- 2- Treat the risk
- 3- Tolerate the risk
- 4- **Transfer the risk**

Question 8

How do we respond to the risk knowing that the probability of facing the risk is low and the cost of the loss is low?

Answers

- 1- Terminate the risk
- 2- Treat the risk
- 3- **Tolerate the risk**
- 4- Transfer the risk

Question 9

We have 2 assets: A and B.

The level of vulnerabilities is 2 for asset A and 3 for asset B,
the level of threats is 3 for asset A and 3 for asset B,
and the level of asset A is 3 and the level of asset B is 1.

In a detailed Risk Analysis approach, which risk should be addressed first?

Answers

- 1- A has an overall risk level of 8 and B has an overall risk level of 9. Asset B should be addressed first because it has higher level of overall risk.
- 2- **A has an overall risk level of 18 and B has an overall risk level of 9. Asset A should be addressed first because it has higher level of overall risk.**
- 3- Both assets should be addressed at the time because they have the same asset level.
- 4- Asset B should be addressed first because the level of vulnerabilities is higher than the level of vulnerabilities of A.

Question 10

We have a company stored data, we need to increase the confidentiality level of this asset, so we used encryption.

Few weeks later we found that using this approach reduced the level of another security concept, which one?

Answers

- 1- Integrity
- 2- **Availability**
- 3- Confidentiality
- 4- None of the above.

Question 11

What is the first step in running a detailed risk analysis approach?

Answers

- 1- **Define the scope**
- 2- Identify the assets and business processes
- 3- Evaluate the assets
- 4- Evaluate the vulnerabilities

Question 12

Using RSA, Alice ciphers her text M with Bob's public key and she gets the ciphertext $C=371$. She sends the ciphertext to Bob who decrypts the ciphertext with his private key according to the following equation $M=C^d \bmod n$ (^ means C powered to d). Bob's public key is $PK=\{e,n\}=\{17,3234\}$ and Bob's private key is $d=2753$. What is the text M that Bob gets? (Use the calculator of your computer in scientific mode).

Answers

- 1- **3185**
- 2- 3234
- 3- 371
- 4- 2703

Question 13

Alice is an IT manager for an international company with branches in many countries, what would be the cost-effective solution for her to interconnect all the branches?

Answers

- 1- Remote Access Server (RAS) with dialup connections
- 2- Leased line connections
- 3- **Virtual Private Network connections**
- 4- Satellite and microwave connections

Question 14

After scanning the open ports on your web server, you discover the ports 22, 23, 25 and 80 are open. Which ports you should close?

Answers

- 1- 22 and 23
- 2- 23, 25 and 80
- 3- 22 and 25
- 4- **23 and 25**

Question 15

What should be the mode of the Network Interface Card (NIC) to allow the sniffer to collect packets destined to all nodes?

Answers

- 1- Prospectus mode
- 2- Catch-all mode
- 3- **Promiscuous mode**
- 4- Heterogeneous mode

Question 16

What is the main difference between worms and viruses?

Answers

- 1- Viruses protect itself against attempt to naturalize it but worms do not.
- 2- Worms work over network, viruses do not.
- 3- **Viruses need a host program to propagate, worms do not.**
- 4- There is no difference between worms and viruses.

Question 17

Which of the following mobile codes present the most level of security threat?

Answers

- 1- Java applet
- 2- JavaScript
- 3- **ActiveX**
- 4- There is no security threat in running mobile code.

Question 18

To detect specific services running on a victim machine, we normally use:

Answers

- 1- **Port Scanning**
- 2- Promiscuous Mode
- 3- Smurf Attack
- 4- Mac Spoofing

Question 19

What is the most critical possible loss resulting from not having VLAN for wireless users in a company?

Answers

- 1- **External users can access inside computers.**
- 2- External users can bring viruses to the network.
- 3- External users can use company services.
- 4- External users can abuse the Internet bandwidth.

Question 20

Ping of death is based upon:

Answers

- 1- **IP fragmentation**
- 2- TCP syn messages
- 3- ICMP reply message
- 4- All of the above

Question 21

Sniffing affects the one of the following:

Answers

- 1- **The confidentiality of the system**
- 2- The availability of the system
- 3- The integrity of the system
- 4- All of the above

Question 22

SYN flooding affects:

Answers

- 1- **The confidentiality of the system**
- 2- The availability of the system
- 3- The integrity of the system
- 4- All of the above

Question 23

Using an anti-virus software eliminates the risk of getting infected by viruses.

Answers

- 1- This is true, if the anti-virus is not hacked.
- 2- This is true, if the anti-virus is used worldwide.
- 3- **This is not true, there is always zero-day risk.**
- 4- This is not true, you need to update the anti-virus periodically.

Question 24

A virus affects in the first place the:

Answers

- 1- The confidentiality of the system
- 2- The availability of the system
- 3- The integrity of the system**
- 4- All of the above

Question 25

Alice wants to check her email using webmail from Bob's computer.
What is the main risk that Alice is taking by doing that?

Answers

- 1- Bob can steal Alice password.**
- 2- Bob can view Alice emails.
- 3- Bob can send emails on Alice behalf.
- 4- There is no risk at all.

Question 26

In a detailed Risk Analysis approach, what is the level of the overall risk if level of vulnerabilities is 2, level of threats is 3 and level of assets is 2?

Answers

- 1- 11
- 2- 12**
- 3- 9
- 4- 7

Question 27

If the probability of facing the risk is high and the cost of the loss is high, how do we respond to the risk?

Answers

- 1- Terminate the risk**
- 2- Treat the risk
- 3- Tolerate the risk
- 4- Transfer the risk

Question 28

Which is not true about using a security policy?

Answers

- 1- Consensus of the system security can be obtained.
- 2- The duty and the responsibility of each person can be defined.
- 3- Security risks that cannot be countered using security tools can be controlled.
- 4- Effective and appropriate security can not be achieved.**

Question 29

Smurf attack is based upon:

Answers

- 1- **IP spoofing and ICMP echo message**
- 2- MAC spoofing
- 3- IP fragmentation
- 4- TCP syn messages

Question 30

DOS attack affects the following:

Answers

- 1- The confidentiality of the system
- 2- **The availability of the system**
- 3- The integrity of the system
- 4- All of the above

Question 31

Deffie-Hellman algorithm is used in the following area:

Answers

- 1- Text encryption and digital signature
- 2- Text encryption only
- 3- **Key exchange only**
- 4- Digital signature and key exchange

Question 32

Callback is one of prevention measures in security:

Answers

- 1- **True**
- 2- False

Question 33

PGP is a:

Answers

- 1- Set of cryptographic routines for digital signatures and message hashing.
- 2- Cryptographic routine for key management
- 3- **Set of cryptographic routines for key management, digital signatures, data encryption and message hashing**
- 4- Set of cryptographic routines for key management and data encryption and message hashing.

Question 34

Alice uses SSL to connect to a web server. What can she benefit from this?

Answers

- 1- Alice can only be sure of the server.
- 2- Alice can be sure of the server, can exchange encrypted data with the server and can sign the messages sent to the server.
- 3- Alice can only exchange encrypted data with the server.
- 4- **Alice can be sure of the server and can exchange encrypted data with the server.**

Question 35

Choose the stream cipher among the following:

Answers

- 1- **RC4**
- 2- RC5
- 3- DES
- 4- MD5

Question 36

Public Key Cryptography solved the problem of key distribution

Answers

- 1- **True**
- 2- False

Question 37

PGP provides a Key Distribution Center which contains all the trusted public keys.

Answers

- 1- True
- 2- **False**

Question 38

Given a hash function H , what do we mean by a collision-resistant for this function?

Answers

- 1- **It should be hard to find two different messages m_1 and m_2 such that $H(m_1) = H(m_2)$**
- 2- Given an input m_1 , it should be hard to find another input, m_2 (not equal to m_1) such that $H(m_1) = H(m_2)$
- 3- Given h it should be hard to find any m such that $h = H(m)$
- 4- Given h it should be hard to find any m such that $m = H(h)$

Question 39

When you receive a digitally signed email then you are sure that:

Answers

- 1- The sender has sent the message and no one else has read the message.
- 2- You only can read the message.
- 3- **The sender has sent the message and that the message has not been altered.**
- 4- The email and that the message has not been altered.

Question 40

The following algorithms are variants of DES algorithm:

Answers

- 1- **Double DES, Triple DES, DESX**
- 2- Double DES, Triple DES, DESX, AES
- 3- Double DES, Triple DES, DESX, IDEA
- 4- Double DES, Triple DES, IDEA

Question 41

Bob encrypted an email with his private key and sent it to Alice and he attached his public key to the email. Which statement from the following is true?

Answers

- 1- Only Alice can read the message.
- 2- No one can read the message, even Alice.
- 3- **Anybody in between can decrypt the message and read it.**
- 4- Anybody in between can decrypt the message and read it except Alice.

Question 42

We have a cipher text C (C1, C2, C3, ...) produced from encrypting a plain text message M of many blocks (M1, M2, M3,...) using the Electronic Code Book (ECB) mode of operation, what are the consequences of decrypting the cipher text if it has an error in the first bit of the first block C1?

Answers

- 1- M1 and the first bit of M2 are wrong
- 2- **Only M1 is wrong**
- 3- M2 and the first bit of M1 are wrong
- 4- Only the first bit of M1 is wrong

Question 43

A digital certificate does not contain:

Answers

- 1- Certificate authority name
- 2- Holder public key
- 3- Holder private key**
- 4- Certificate usage domain

Question 44

RSA algorithm is used in the following areas:

Answers

- 1- Digital signature and key exchange**
- 2- Key exchange only
- 3- Text encryption and digital signature
- 4- Text encryption only

Question 45

Bob encrypted an email with his private key and sent it to Alice and he attached his public key to the email. Which statement from the following is true?

Answers

- 1- Only Alice can read the message.
- 2- No one can read the message, even Alice.
- 3- Anybody in between can decrypt the message and read it.**
- 4- Anybody in between can decrypt the message and read it except Alice.

Question 46

We have a cipher text C ($C_1, C_2, C_3, \dots$) produced from encrypting a plain text message M of many blocks ($M_1, M_2, M_3, \dots$) using the Output Feedback (OFB) mode of operation, what are the consequences of decrypting the cipher text if it has an error in the first bit of the first block C_1 ?

Answers

- 1- Only M_1 is wrong
- 2- Only the first bit of M_1 is wrong**
- 3- M_2 and the first bit of M_1 are wrong
- 4- M_1 and the first bit of M_2 are wrong

Question 47

You have a company with branches distributed in many cities, what would be the safest solution to interconnect all the branches?

Answers

- 1- **Leased line connections**
- 2- Remote Access Server (RAS) with dialup connections
- 3- Virtual Private Network connections
- 4- Satellite connection

Question 48

DES algorithm key size is:

Answers

- 1- 128 bits
- 2- 48 bits
- 3- **56 bits**
- 4- 64 bits

Question 49

In symmetric cryptography :

Answers

- 1- We use half-key from the sender and half-key from the receiver.
- 2- **We use the same key for encryption and decryption.**
- 3- We use one pair of keys.
- 4- We do not use keys, it depends on keeping the algorithm unknown.

Question 50

Public key cryptography consists of:

Answers

- 1- Two keys, neither of which should be published
- 2- One key which is shared between sender and receiver
- 3- One key which is transmitted via a separate method from the data.
- 4- **Two keys, one of which is published for anyone's use**

Question 51

More advanced firewall products implement this protection technique:

Answers

- 1- **Stateful inspection.**
- 2- Network Address Translation (NAT)

- 3- Spoofing Protocols
- 4- Packet filtering

Question 52

Alice wants to check her email using webmail from Bobs computer.
What is the main risk that Alice is taking by doing that?

Answers

- 1- Bob can steal Alice password.**
- 2- Bob can view Alice emails.
- 3- Bob can send emails on Alice behalf.
- 4- There is no risk at all.

Question 53

DESX algorithm uses a key whose length is:

Answers

- 1- 56 bits
- 2- 64 bits
- 3- 120 bits**
- 4- 128 bits

Question 54

Which of the following should not be included in the security policy document?

Answers

- 1- What users cannot do.
- 2- Rules of user authentication.
- 3- Penalties for inappropriate behaviors.
- 4- None of the above.**

Question 55

Deffie-Hellman algorithm is used in the following area:

Answers

- 1- Text encryption and digital signature
- 2- Text encryption only
- 3- Key exchange only**
- 4- Digital signature and key exchange

Question 56

Describe the process of user to server authentication using Kerberos:

Answers

Question 57

Describe the process of generating digital signature and give examples of the required algorithms, then explain how the receiver can verify the signature.

Answers

Question 58

Explain the difference between Stream Cipher and Block Cipher and give some types of these techniques and name some algorithms of each one.

Answers

Question 59

Alice has a pair of RSA keys, the public key is (n,e) , the clear text message M and the cipher text $C=M^e \pmod n$.

We suppose that Bob wants to encrypt two successive messages, and send them to Alice, the messages are of the form, M and $M+w$, the corresponding ciphers are C and C' .

We suppose that an attacker intercepted C and C' and that he knows w . Find out how can the attacker calculate the message M , in the case of $e=2$.

Answers

Question 60

Alice has a pair of RSA keys, the public key is (n, e) , the clear text message M and the cipher text $C = M^e \pmod{n}$.

We suppose that Bob wants to encrypt two successive messages, and send them to Alice, the messages are of the form, M and $M+w$, the corresponding ciphers are C and C' .

We suppose that an attacker intercepted C and C' and that he knows w . Find out how can the attacker calculate the message M , in the case of $e=2$.

Answers

Question 61

Explain the role of the firewall and describe the difference between stateful and stateless inspection.

Answers

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